Forecasting How Many Loaves to Bake Each Morning

Summary

- **Problem:** the bakery bakes based on sales from the same period last year, so it ends up with both leftovers and shortages. A shortage costs about 3 times more than a leftover.
- **Approach:** train AI on 2021 data, run an exam on January–June 2022, then a live run on July–September 2022, and compare against the current manual method.
- **Results:** shortages fell from 13.8% to 5.7% in the exam and from 11.4% to 8.5% in the live run. Total money lost fell by 28% and 17%.
- **Recommendation:** not worth it for a single shop, which takes about 3 years to pay back; for a chain, pilot it in 2–3 shops.

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1. The customer's problem

Hypothetical scenario based on public sales data from a bakery in France (Kaggle).

The owner needs to forecast how many of each type of bread to bake every morning.

Today, the bakery bakes based on **sales from the same period last year**: whatever sold on the same weekday, in the same week last year, is what gets baked. This keeps up with the tourist season because the seasonal pattern repeats, but it goes wrong when this year differs from last year. Baking the wrong quantity costs money in two ways: bake too many and the leftovers are thrown away; bake too few and customers arrive to find nothing left. So the owner is looking for a better way to forecast.

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2. What the customer's data shows

2.1. Data provided by the customer

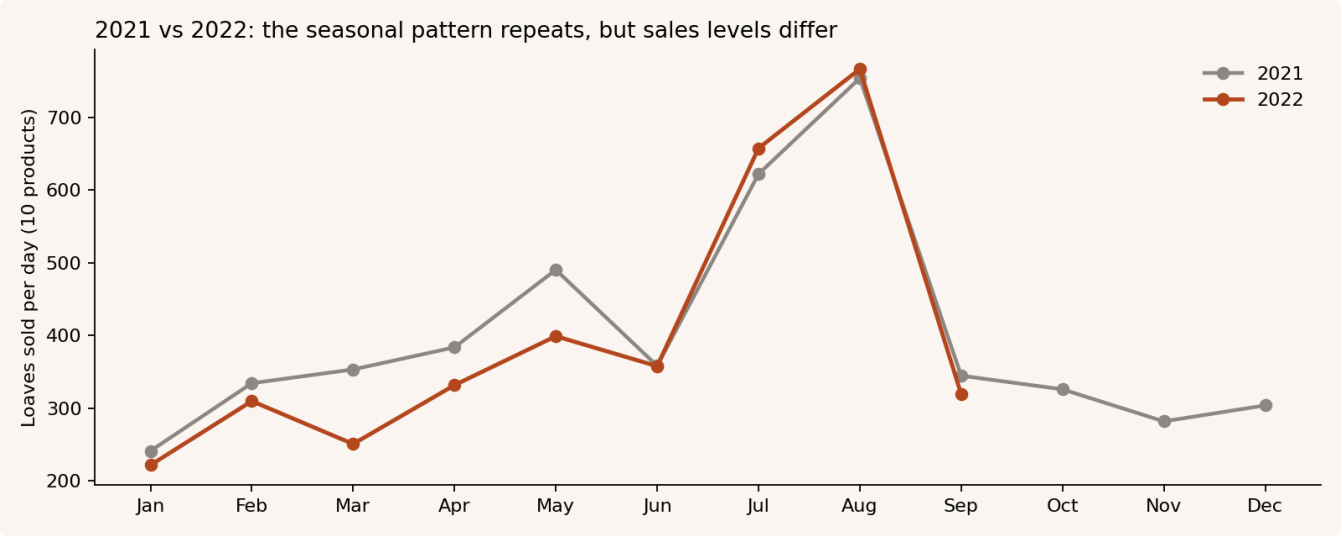
Data	Content	Used for
Daily sales	Units sold of each product per day	Building the plan (using sales **before** that day) and scoring it (sales **on** that day)
Recipes	Kilograms of each ingredient per unit	Calculating the cost of one unit
Ingredient prices	Price per kilogram	Calculating the cost of one unit
Selling prices	Price of each product	Calculating the margin on one unit

Data	Content	Used for
Closed days	Days the bakery was closed	Removing closed days

Not in the data: units actually baked each day, leftovers, and transaction times. So we don't know how much the bakery **actually** over- or under-baked. Leftovers and shortages below are calculated as described in "How leftovers and shortages are calculated".

2.2. Same period, 2021 vs 2022

This chart shows what the sales data says: average units sold per day (the 10 best-selling products), by month, with 2021 next to 2022.



Both years share the same shape (July–August is twice as busy), but they are not identical: March–May 2022 sold noticeably less than the year before.

2.3. How leftovers and shortages are calculated

Since we don't have the units actually baked, we set a **baking plan** for each day and compare it with **what customers actually bought** that day:

- The plan is calculated **only from sales before that day**, just as nobody knows that day's sales when baking in the morning.
- If the plan is **higher** than what was bought, the difference is a **leftover**. If it is **lower**, the difference is a **shortage**.

2.4. From leftovers and shortages to money lost

Using only the customer's data:

	How it's calculated	Traditional baguette
One leftover loses	Cost: ingredient cost of one unit	€0.18
One shortage loses	Margin: selling price − ingredient cost	€1.02

Money lost = leftovers × cost + shortages × margin, calculated per product and then summed.

2.5. Baking the same as last year: results by month

The plan for each day is the sales on **the same weekday, exactly 52 weeks earlier** in 2021 (the bakery's current manual method). Leftovers, shortages and money lost are calculated per day and per product, then **summed by month**. Figures cover 10 products.

Month 2022	Planned	Customers bought	Leftovers	Shortages	Money lost
Jan	6,051	5,768	1,030	747	€1,000
Feb	8,267	7,436	1,653	822	€1,239
Mar	9,031	6,940	2,446	355	€894
Apr	11,356	9,797	2,934	1,375	€2,024
May	12,715	11,056	4,111	2,452	€3,346
Jun	10,778	10,736	1,416	1,374	€1,735
Jul	19,959	20,379	1,958	2,378	€2,859
Aug	22,709	23,756	1,692	2,739	€3,093
Sep	9,586	8,946	1,558	918	€1,343
Total	**110,452**	**104,814**	**18,798**	**13,160**	**€17,533**

Over 9 months, baking the same as last year lost about **€17,500**, around **75% of it from shortages**. May lost the most: 2022 sold less than 2021, so there were many leftovers, yet still many days with shortages.

*The data has no labour or energy costs, so the money lost to leftovers here is a **minimum**. The money lost to shortages is also a minimum, because the shortage count is a minimum.*

2.6. Do leftovers or shortages cost more?

The logic: take a loaf that sells for 10, costs 3 to make and earns a margin of 7:

Case	Cost spent	Revenue	Lost
Leftover: one baked, nobody bought it	3	0	3, i.e. the **cost**
Shortage: a customer wanted one but none were left	0	0	7, i.e. the **margin** that could have been earned

With a shortage, the loaf was never baked, so no cost was spent; what is lost is the margin. So: **if the margin is larger than the cost, shortages hurt more**; if the cost is larger than the margin, leftovers hurt more.

What the data shows: for all 10 products, the margin is 2 to 6.5 times the cost (details by product in Section 3).

Applied to the bakery's current method (January–September 2022):

	Units	Money lost
Leftovers	18,798	about €4,400
Shortages	13,160	about €13,100

Conclusion: there are fewer shortage units than leftover units, but shortages cost **about 3 times as much**. The bakery's biggest problem is **shortages**, not leftovers.

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3. Approach and scoring

3.1. Goal

Reduce shortages. Section 2 shows that shortages cost about 3 times more than leftovers, so this is where improvement matters most.

But there is a trap: the easiest way to eliminate shortages is to bake a lot. For example, bake three times as much and you will never run out, but the cost sunk into leftovers will far exceed the margin you save. So there is one constraint: **total money lost (leftovers plus shortages) must be lower than with the manual method.**

3.2. Three phases

The data is split by time, like taking an exam before starting the real job:

Phase	Period	Used for
Learning	January–December 2021	AI learns the patterns. The manual method takes its same-period figures from here
Exam	January–June 2022	Compare AI with the manual method, then decide whether to go live
Live run	July–September 2022	Simulate real use. Everything is locked after the exam; **nothing changes**

This split is also a tough test for AI. The exam falls in **spring**, when 2022 differs from 2021. The live run falls in the **tourist season**, when the manual method does best because the seasonal pattern repeats.

3.3. Two forecasting methods

	Manual forecast	AI forecast
What it is	The bakery's current method	A machine learning model
How	Bake the sales of **the same weekday, exactly 52 weeks earlier** in 2021	Learns from sales data, then forecasts each day
Based on	A single day from last year	Patterns of weekday, season and holidays

Why could AI beat the manual method? The manual method takes exactly one day from last year, so if that day was unusual, the plan is off too. AI learns from the whole year, so it can separate weekday, seasonal and holiday patterns.

How is AI used day to day? Every evening, AI suggests how many of each product to bake. The owner approves or adjusts the numbers before baking. AI does not decide on its own.

3.4. How AI prioritises fewer shortages

If AI simply predicted the middle, it would run short on about **half of all days**. Because shortages hurt more than leftovers, AI is set **above the middle**: bake enough to avoid shortages on about **4 out of 5 days**. Here is why 4 out of 5.

Think about the last extra loaf. There are two possibilities:

- A customer buys it: the bakery **keeps the margin**.
- Nobody buys it: the bakery **loses the cost**.

Baking that loaf is worth it if the margin kept outweighs the cost at risk. The balance point is:

> **Target in-stock rate = margin ÷ (margin + cost)**

Example: a loaf sells for 10, costs 3, margin 7: target in-stock rate = 7 ÷ (7 + 3) = **70%**. That means baking enough that about 7 out of 10 days have no shortage. Bake less and you run short too often, losing margin. Bake more and those extra loaves rarely sell, losing cost.

Applied to the bakery's 10 products (margin and cost from Section 2):

Product	Margin	Cost	Target in-stock rate
Pain au chocolat	€0.81	€0.39	67%
Cereal baguette	€0.93	€0.32	74%
Croissant	€0.86	€0.24	78%
Baguette	€0.72	€0.18	80%
Special bread	€1.93	€0.47	80%
Boule 400g	€1.22	€0.28	81%
Campagne	€1.48	€0.32	82%
Banette	€0.87	€0.18	83%
Traditional baguette	€1.02	€0.18	85%
Tartelette	€1.73	€0.27	87%

The middle value across the 10 products is about **81%**, roughly **4 out of 5 days**. AI uses this common rate for all 10 products. It is calculated from margin and cost, not tuned to the 2022 results.

3.5. Scoring

Both AI's plan and the manual plan are **compared with what customers actually bought** in 2022, day by day and product by product, as described in Section 2. The data has no record of the bakery's actual losses, so this is the only way to compare the two plans.

Metric	How it's calculated	Role
Shortages	Shortage units ÷ units sold	**Primary metric:** the goal is to reduce this

Metric	How it's calculated	Role
Total money lost	Leftovers × cost + shortages × margin	**Constraint:** must be lower than with the manual method
Leftovers	Leftover units ÷ units sold	Monitor: how many leftovers it takes to reduce shortages

Beyond the totals, look at results by month and by product to see where AI does better and where it does worse.

3.6. Pass criteria and decision rules

AI **passes** when, compared with the manual method, it has **fewer shortages** and **lower total money lost**. The pass criteria are written before running and do not change after seeing the results.

- **After the exam:** pass, move to the live run; fail, stop and find the cause.
- **After the live run:** pass, move on to costing; fail, do not scale up, find the cause and fix it.

Passing does not mean deploying. AI costs money to build and run. Whether to use it is decided in the next step, by comparing **the money saved** with **the cost of AI**.

3.7. Expected outcome

Summarising the sections above:

Primary goal	Fewer shortages than the manual method
Constraint	Lower total money lost than the manual method
Accepted trade-off	Slightly more leftovers, in exchange for fewer shortages
Instruction to AI	Bake enough for about 4 out of 5 days (81% rate)
Must hold in	Both the exam (January–June 2022) and the live run (July–September 2022)

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4. Prototype

The prototype goes through four steps: preparing the data, training AI, taking the exam, then the live run. I built it in **Google Colab** (Python in the browser), with **Claude helping write the code**. My part was the problem framing, the data used, how AI is instructed and how it is scored; choosing the model type is the technical part. The result tables and charts in this section come from that notebook.

4.1. Data preparation and cleaning

Cleaned data: each row is one day of one product, with units sold, cost, margin, holiday flag and manual plan. **6,000 rows** in total (10 products × 600 open days).

The prototype uses only **the 10 best-selling products among those with recipes**, because 101 of the 148 products have no recipe, so their cost cannot be calculated.

Cleaned data: 6,000 rows (10 products × 600 open days), 10 sample rows

Date	Product	Sold	Cost (€)	Margin (€)	Holiday	Manual plan
2022-02-07	TRADITIONAL BAGUETTE	101	0.18	1.02	No	106
2022-02-07	CROISSANT	13	0.24	0.86	No	28
2022-02-07	PAIN AU CHOCOLAT	14	0.39	0.81	No	14
2022-02-07	BANETTE	38	0.18	0.87	No	13
2022-02-07	BAGUETTE	27	0.18	0.72	No	22
2022-02-07	CEREAL BAGUETTE	9	0.32	0.93	No	2
2022-02-07	SPECIAL BREAD	5	0.47	1.93	No	8
2022-02-07	TARTELETTE	6	0.27	1.73	No	13
2022-02-07	BOULE 400G	6	0.28	1.22	No	7
2022-02-07	CAMPAGNE	2	0.32	1.48	No	2

Each row is one day of one product.

4.2. Training AI

AI learns from past sales to find patterns: how much of each product sells, by weekday, by season and on holidays. The output is **how many of each product to bake each day**.

AI is instructed to plan for a **target in-stock rate of 81%** (Section 3), that is, to prioritise fewer shortages.

AI is trained twice:

	Run 1: for the exam	Run 2: for the live run
Learns from	2021	2021 and January–June 2022

	Run 1: for the exam	Run 2: for the live run
Forecasts	January–June 2022	July–September 2022

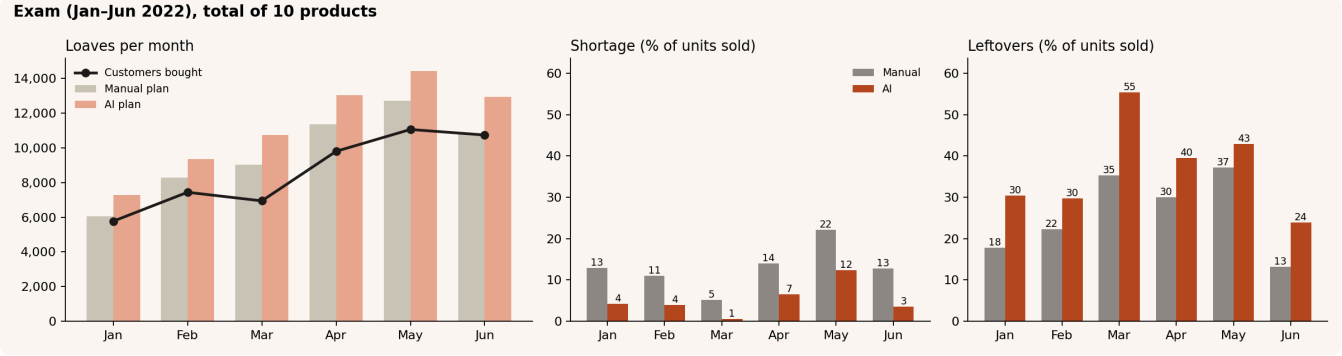
Training complete. Target in-stock rate: 81%. Sample plan for 2022-01-08

Date	Product	Manual plan	AI plan	Customers bought
2022-01-08	TRADITIONAL BAGUETTE	135	135	84
2022-01-08	CROISSANT	33	41	22
2022-01-08	PAIN AU CHOCOLAT	25	27	19
2022-01-08	BANETTE	30	29	20
2022-01-08	BAGUETTE	25	30	20
2022-01-08	CEREAL BAGUETTE	10	10	9
2022-01-08	SPECIAL BREAD	8	8	4
2022-01-08	TARTELETTE	0	5	8
2022-01-08	BOULE 400G	6	6	6
2022-01-08	CAMPAGNE	5	5	2

Run 1 learns from 2021, forecasts Jan–Jun 2022. Run 2 learns from 2021 + Jan–Jun 2022, forecasts Jul–Sep 2022.

4.3. Exam (January–June 2022)

By month, total of 10 products:



Exam results (Jan–Jun 2022): PASS

	Manual	AI
Customers bought (units)	51,733	51,733
Planned (units)	58,198	67,767
Shortage (units)	**7,125**	**2,957**
Shortage (%)	**13.8%**	**5.7%**
Lost to shortage	€7,117	€2,917
Leftover (units)	13,590	18,991
Leftover (%)	26.3%	36.7%
Lost to leftovers	€3,122	€4,435
Total money lost	**€10,239**	**€7,353**

Results for all 6 months:

	Manual	AI
Customers bought	51,733 units	51,733 units
Planned	58,198 units	67,767 units
Shortages	**7,125 units (13.8%)**	**2,957 units (5.7%)**
Lost to shortages (margin)	€7,117	€2,917
Leftovers	**13,590 units (26.3%)**	**18,991 units (36.7%)**
Lost to leftovers (cost)	€3,122	€4,435
Total money lost	**€10,239**	**€7,353**

Percentages are based on the units customers bought.

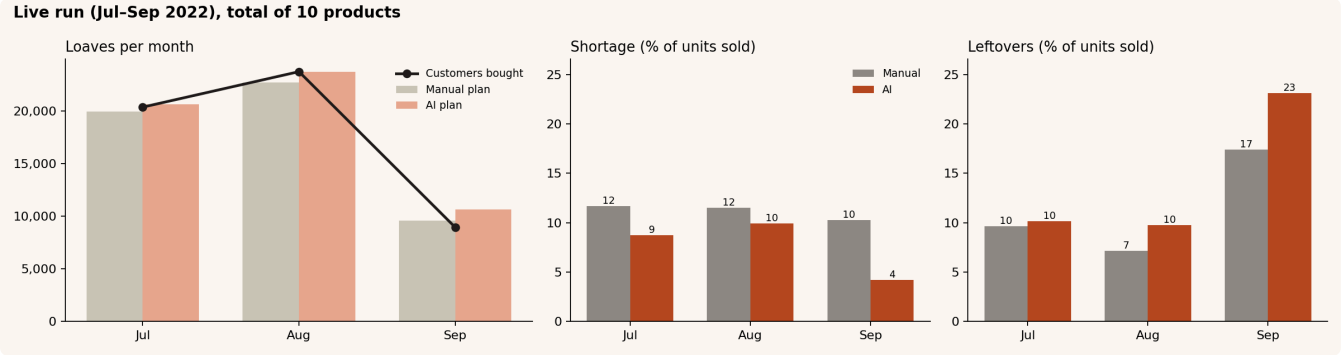
- Shortages fell by **more than half**, from 13.8% to 5.7%.

- Total money lost fell by **28%**.
- In exchange, leftovers rose from 26% to 37%.
- Summed by month, both plans baked more than customers bought, yet there were still shortages. That is because shortages happen day by day, product by product: today's leftovers cannot serve a customer who misses out on another day.

Decision: pass. Move AI to the live run, keeping all settings unchanged.

4.4. Live run (July–September 2022)

By month, total of 10 products:



Live run results (Jul–Sep 2022): PASS

	Manual	AI
Customers bought (units)	53,081	53,081
Planned (units)	52,254	55,016
Shortage (units)	**6,035**	**4,518**
Shortage (%)	**11.4%**	**8.5%**
Lost to shortage	€6,024	€4,454
Leftover (units)	5,208	6,453
Leftover (%)	9.8%	12.2%
Lost to leftovers	€1,271	€1,598
Total money lost	**€7,295**	**€6,052**

Results for all 3 months:

	Manual	AI
Customers bought	53,081 units	53,081 units
Planned	52,254 units	55,016 units
Shortages	**6,035 units (11.4%)**	**4,518 units (8.5%)**
Lost to shortages (margin)	€6,024	€4,454
Leftovers	**5,208 units (9.8%)**	**6,453 units (12.2%)**
Lost to leftovers (cost)	€1,271	€1,598
Total money lost	**€7,295**	**€6,052**

- The tourist season repeated last year’s pattern, so the manual method did fairly well.

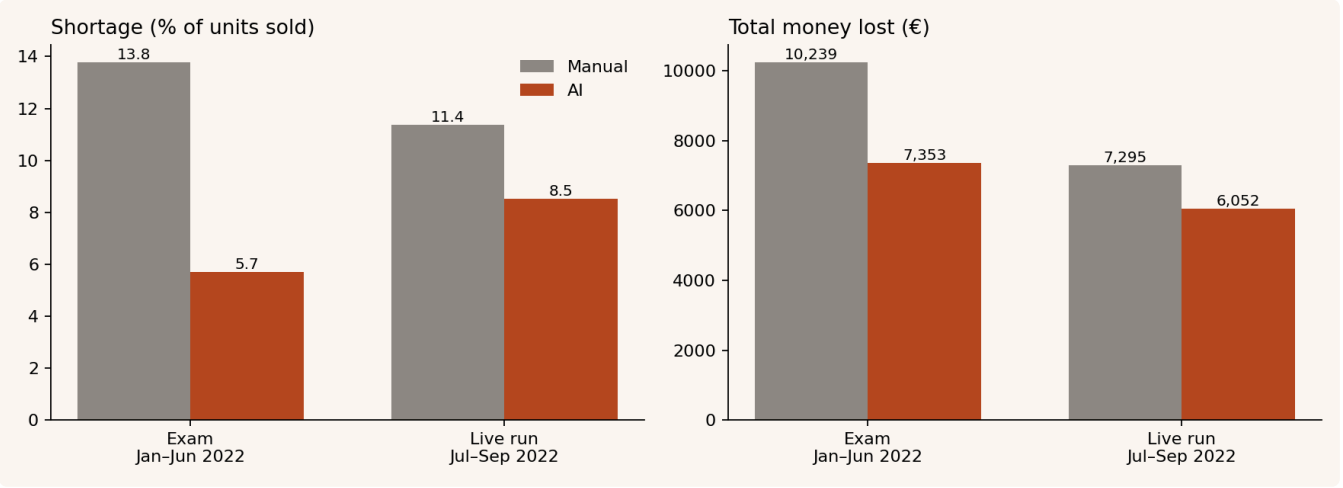
- AI still had fewer shortages, total money lost fell by **17%**, and leftovers rose only slightly.

Decision: pass. Move on to costing to decide whether to deploy.

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5. Decision and next steps

5.1. Looking back at both phases



AI **passed both the exam and the live run**: fewer shortages and lower total money lost than the manual method in both phases.

5.2. How much AI saves

January–September 2022, 10 products	Manual	AI
Shortages	13,160 units	7,475 units
Leftovers	18,798 units	25,444 units
Total money lost	**€17,534**	**€13,405**

AI reduced losses by about **€4,100 over 9 months**, or about **€460 per month** for one shop.

5.3. Estimated cost and payback period (assumptions)

Detailed figures will be estimated by the AI team before approval; the project plan is drawn up after the deployment decision.

Item	Who	People and duration	Cost
Build the recommendation system	Data scientist	1 person, 20 days × €400	€8,000 (one-off)
Connect daily sales data	Data engineer	1 person, 3 days × €400	€1,200 (one-off)
Project management, working with the owner, scoring results	AI PM	1 person, 5 days × €400	€2,000 (one-off)
Server running the model each evening	Cloud service	No staff needed	€30 / month
Updating and monitoring the model	Data scientist	1 person, 1 day per quarter	about €130 / month
Total			**€11,200 one-off, about €160 / month**

Compared with the savings:

	One shop	Chain of 10 shops
One-off cost	€11,200	about €14,000
Monthly cost	about €160	about €180
Monthly savings	about €460	about €4,600
Net gain per month	**about €300**	**about €4,420**
Payback period	**about 3 years**	**about 3 months**

The data covers only one shop. The "Chain of 10 shops" column is a hypothetical scenario, to show how the result changes when the system is used across many shops. The one-off cost of about €14,000 consists of:

- *Building the recommendation system: €8,000. Built once, with one model shared by all 10 shops.*
- *Connecting sales data: €2,000 (5 days, 2 more than for one shop). This assumes all 10 shops use the same point-of-sale system; if each uses a different one, this cost rises significantly.*
- *AI PM: €4,000 (10 days), to work with several owners and score results for each shop.*

Monthly costs are €50 for the server and about €130 for model updates. This assumes each shop saves as much as the shop in the data.

Conclusion: for **one shop**, AI is barely worth it, as it takes about 3 years to pay back. For **a chain of shops**, AI is worth it, because the build cost is paid once while the savings multiply with the number of shops.

5.4. What to confirm before deciding

- **What a leftover loaf really costs.** The €4,100 figure counts only ingredient cost for leftovers. If labour is added at 20% of the selling price, the live-run result reverses: AI does about 13% worse than the manual method.
- **Whether the owner accepts more leftovers.** AI reduces shortages by baking more, producing about 6,600 more leftover units over 9 months.

5.5. Recommendation

- **One shop:** not yet, since payback takes about 3 years.
- **Chain of shops:** pilot in 2–3 shops for 8 weeks, after confirming the two points above. During the pilot, record **units baked and leftovers** each day, to measure leftovers directly rather than estimate them.
- **The final decision** rests with the owner or management, based on costs confirmed by the AI team.